Business Engineering – Exercise Sheet: Vectors, Matrices, Tensors, and Eigenanalysis in Logistics

Bachelor in Business Engineering

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Exercises on Vectors, Matrices, Tensors, Eigenvalues, and Diagonalization

Use exact expressions whenever convenient and round numerical answers to three decimals when needed. Unless otherwise stated, vectors are column vectors and matrices act from the left:

$$y = Ax.$$

In logistics interpretations, use the convention that columns represent source objects and rows represent target objects unless explicitly stated otherwise. Thus, the entry a_{ij} describes the weight, cost, flow fraction, or relation from source j to target i.
When useful, use

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n, \qquad A = (a_{ij}) \in \mathbb{R}^{m\times n}, \qquad y = Ax, \qquad y_i = \sum_{j=1}^{n} a_{ij}x_j,$$

$$c^T x = \sum_{i=1}^{n} c_i x_i, \qquad (BA)_{ik} = \sum_{j=1}^{m} b_{ij}a_{jk}, \qquad A^T = (a_{ji}),$$

$$\langle C, X\rangle = \sum_i \sum_j c_{ij}x_{ij}, \qquad Ax = b,$$

$$Av = \lambda v, \qquad w^T A = \lambda w^T, \qquad A^T w = \lambda w,$$

$$\det(A - \lambda I) = 0, \qquad \chi_A(\lambda) = \lambda^2 - \operatorname{tr}(A)\lambda + \det(A) \quad \text{for } A \in \mathbb{R}^{2\times 2},$$

$$A = PDP^{-1}, \qquad A^k = PD^kP^{-1}.$$

Part A – Vectors, matrices, tensors, and logistics models (Exercises 1–10)

1. **Ex 1: Scalars, vectors, matrices, and tensors in logistics**

   Consider the following objects:

$$9.5, \qquad s = \begin{pmatrix} 140 \\ 90 \\ 75 \\ 110 \end{pmatrix}, \qquad C = \begin{pmatrix} 4 & 7 & 5 \\ 6 & 3 & 8 \end{pmatrix},$$

and

$$X_{pijt}, \qquad p = 1, \ldots, 3, \quad i = 1, \ldots, 2, \quad j = 1, \ldots, 4, \quad t = 1, \ldots, 5.$$

(a) Classify each object as scalar, vector, matrix, or tensor. Give the dimensions whenever applicable. (6 Points)

(b) Give one logistics interpretation for s, one for C, and one for X_{pijt}. (6 Points)

(c) Determine the total number of scalar entries contained in s, C, and X. (4 Points)

(d) Interpret the statement

$$X_{2,1,3,4} = 18$$

in one precise logistics sentence. (4 Points)

2. **Ex 2: Demand, stock, shortage, and cost vectors**

Three stores have current stock

$$s = \begin{pmatrix} 95 \\ 90 \\ 70 \end{pmatrix},$$

expected demand

$$d = \begin{pmatrix} 120 \\ 80 \\ 100 \end{pmatrix},$$

and required safety stock

$$r = \begin{pmatrix} 10 \\ 10 \\ 15 \end{pmatrix}.$$

The emergency procurement cost vector is

$$c = \begin{pmatrix} 4 \\ 5 \\ 3 \end{pmatrix} \quad \text{EUR/unit.}$$

(a) Compute the demand-minus-stock vector $d - s$. (4 Points)

(b) Compute the required shortage vector

$$h = d + r - s.$$

Then replace all negative components by zero, because negative shortages mean surplus stock. (6 Points)

(c) Compute the total emergency procurement cost

$$c^T h.$$

(6 Points)

(d) Suppose demand increases by 20%. Write the new demand vector as a scalar multiple of d, and explain the meaning of scalar multiplication in this context. (4 Points)

3. **Ex 3: Matrix-vector multiplication as supplier-to-hub allocation**

Three suppliers provide quantities

$$q = \begin{pmatrix} 50 \\ 40 \\ 60 \end{pmatrix}.$$

The allocation matrix from suppliers to two hubs is

$$A = \begin{pmatrix} 0.6 & 0.3 & 0.2 \\ 0.4 & 0.7 & 0.8 \end{pmatrix}.$$

Rows represent hubs and columns represent suppliers.

(a) State the dimensions of A, q, and Aq. (4 Points)

(b) Compute the hub inflow vector

$$h = Aq.$$

(8 Points)

(c) Check the column sums of A. What do they imply about conservation of total quantity? (4 Points)

(d) Interpret the entry

$$a_{23} = 0.8$$

in one logistics sentence. (4 Points)

4. **Ex 4: Matrix-matrix multiplication as composition of logistics relations**

A supplier-to-hub allocation matrix is

$$A = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix},$$

where rows are hubs and columns are suppliers.

A hub-to-store allocation matrix is

$$B = \begin{pmatrix} 0.6 & 0.1 \\ 0.4 & 0.9 \end{pmatrix},$$

where rows are stores and columns are hubs.

(a) Compute the direct supplier-to-store matrix

$$BA.$$

(8 Points)

(b) Interpret the entry

$$(BA)_{21}$$

in one logistics sentence. (4 Points)

(c) For supplier quantity vector

$$q = \begin{pmatrix} 100 \\ 50 \end{pmatrix},$$

compute the final store inflow vector

$$BAq.$$

(4 Points)

(d) Explain why BA represents composition of two logistics relations rather than ordinary entrywise multiplication. (4 Points)

5. **Ex 5: Non-symmetric road-time matrices**

In this exercise only, let T_{ij} denote the road travel time from origin i to destination j. Consider

$$T = \begin{pmatrix} 0 & 18 & 25 \\ 22 & 0 & 14 \\ 20 & 16 & 0 \end{pmatrix} \quad \text{minutes.}$$

(a) Compute the transpose T^T. (4 Points)

(b) Determine whether T is symmetric. Identify all origin-destination pairs for which the travel time differs by direction. (6 Points)

(c) Compute the round-trip matrix

$$R = T + T^T.$$

Which two-location round trip is shortest? (6 Points)

(d) Explain why Euclidean straight-line distance is symmetric, but real road travel time is often not symmetric. (4 Points)

6. **Ex 6: Shipment matrices and total transport cost**

A company ships from two warehouses to three stores. The shipment matrix is

$$X = \begin{pmatrix} 20 & 30 & 10 \\ 40 & 10 & 20 \end{pmatrix},$$

where rows are warehouses and columns are stores. The route cost matrix is

$$C = \begin{pmatrix} 5 & 6 & 4 \\ 7 & 3 & 8 \end{pmatrix} \quad \text{EUR/unit.}$$

(a) Compute the row sums and column sums of X. Interpret both. (6 Points)

(b) Compute the total transport cost

$$\sum_{i=1}^{2}\sum_{j=1}^{3} c_{ij}x_{ij}.$$

(6 Points)

(c) Compute the total cost generated by warehouse 1 and the total cost generated by store 3. (4 Points)

(d) Write the total cost compactly as a matrix inner product

$$\langle C, X\rangle.$$

Explain why this is not ordinary matrix multiplication. (4 Points)

7. **Ex 7: Solving a vehicle-planning problem with $Ax = b$**

A logistics planner must choose the numbers of three vehicle types:

$$x = \text{small vehicles}, \qquad y = \text{medium vehicles}, \qquad z = \text{large vehicles}.$$

The planning constraints are:

$$2x + 4y + 8z = 68 \quad \text{pallet capacity},$$
$$x + y + z = 15 \quad \text{number of vehicles},$$
$$x + 2y + 3z = 30 \quad \text{driver-hour units}.$$

(a) Write the system in matrix form

$$Ax = b.$$

(4 Points)

(b) Solve the system using Gaussian elimination. (8 Points)

(c) Verify that the solution satisfies all three original equations. (4 Points)

(d) Explain what a negative or non-integer solution would mean in this logistics planning context. (4 Points)

8. **Ex 8: Incidence matrices and network-flow balance**

A small logistics network has three nodes:

$$1 = \text{supplier}, \qquad 2 = \text{hub}, \qquad 3 = \text{store}.$$

The arcs are

$$x_1 : 1 \to 2, \qquad x_2 : 1 \to 3, \qquad x_3 : 2 \to 3.$$

Use the incidence matrix

$$A = \begin{pmatrix} -1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{pmatrix}.$$

Positive entries in $b = Ax$ mean net inflow, and negative entries mean net outflow.

(a) For

$$x = \begin{pmatrix} 30 \\ 20 \\ 25 \end{pmatrix},$$

compute

$$b = Ax.$$

(6 Points)

(b) Interpret each component of b as supply, demand, or transshipment balance. (6 Points)

(c) Given the balance vector

$$b = \begin{pmatrix} -60 \\ 10 \\ 50 \end{pmatrix}$$

and the direct supplier-to-store flow

$$x_2 = 25,$$

solve for x_1 and x_3. (6 Points)

(d) Explain why a necessary condition for feasibility is

$$b_1 + b_2 + b_3 = 0.$$

(2 Points)

9. **Ex 9: Tensors for product-location-time logistics data**

Let

$$X_{pijt}$$

denote the quantity of product p shipped from warehouse i to store j on day t, where

$$p = 1, \ldots, 2, \qquad i = 1, \ldots, 3, \qquad j = 1, \ldots, 4, \qquad t = 1, \ldots, 7.$$

(a) State the order and dimensions of the tensor X. How many scalar entries does it contain? (5 Points)

(b) Define

$$Y_{ij} = \sum_{p=1}^{2} \sum_{t=1}^{7} X_{pijt}.$$

What are the dimensions of Y, and what does Y_{ij} mean? (5 Points)

(c) For store $j = 2$, define a product-demand vector

$$d_p = \sum_{i=1}^{3} \sum_{t=1}^{7} X_{pi2t}.$$

What is the dimension of d, and what does each component mean? (5 Points)

(d) For fixed $p = 1$ and $t = 3$, explain what the matrix

$$X_{1ij3}$$

represents. (5 Points)

10. **Ex 10: Matrices as linear maps and weighted relations**

Consider the matrix

$$A = \begin{pmatrix} 0.5 & 0.1 & 0 \\ 0.5 & 0.7 & 0.2 \\ 0 & 0.2 & 0.8 \end{pmatrix}.$$

Let e_1, e_2, e_3 be the standard basis vectors of $\mathbb{R}^3$.

(a) Compute

$$Ae_1, \qquad Ae_2, \qquad Ae_3.$$

Interpret each column as a weighted logistics relation from one source node to three target nodes. (6 Points)

(b) Let

$$x = \begin{pmatrix} 10 \\ 0 \\ 5 \end{pmatrix}, \qquad y = \begin{pmatrix} 0 \\ 20 \\ 0 \end{pmatrix}.$$

Verify numerically that

$$A(x + y) = Ax + Ay.$$

(6 Points)

(c) Consider the permutation matrix

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Compute

$$P \begin{pmatrix} W_1 \\ W_2 \\ W_3 \end{pmatrix}$$

and interpret the result. (4 Points)

(d) Explain why the set of invertible 3×3 matrices forms a group under matrix multiplication. Mention the identity element and the inverse element. (4 Points)

Part B – Eigenvalues, eigenvectors, diagonalization, and repeated logistics processes (Exercises 11–20)

11. **Ex 11: Checking eigenvectors directly**

Consider

$$D = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$$

and

$$M = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}.$$

(a) Check whether

$$e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad \text{and} \qquad e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

are eigenvectors of D. Determine the corresponding eigenvalues. (6 Points)

(b) Check whether

$$v = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

is an eigenvector of M. Determine the eigenvalue if it is one. (4 Points)

(c) Check whether

$$u = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

is an eigenvector of M. Determine the eigenvalue if it is one. (6 Points)

(d) Explain in one sentence why the zero vector is not allowed to be called an eigenvector. (4 Points)

12. **Ex 12: Characteristic equation of a triangular logistics matrix**

Consider

$$B = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}.$$

(a) Compute

$$\det(B - \lambda I).$$

(5 Points)

(b) Determine all eigenvalues of B. (5 Points)

(c) Compute one right eigenvector for each eigenvalue. (6 Points)

(d) Suppose B is applied repeatedly to a demand-pattern vector. Which eigenvalue determines the dominant long-run growth mode? Explain briefly. (4 Points)

13. **Ex 13: Right eigenvectors of a two-hub redistribution process**

Every night, stock is redistributed between two hubs according to

$$x_{k+1} = Ax_k, \qquad A = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}.$$

Columns represent source hubs and rows represent target hubs.

(a) Verify that the columns of A sum to 1. What does this imply operationally? (4 Points)

(b) Compute the characteristic polynomial

$$\det(A - \lambda I)$$

and determine the eigenvalues. (6 Points)

(c) Compute one right eigenvector for each eigenvalue. (6 Points)

(d) Interpret the eigenvector corresponding to $\lambda = 1$ and the eigenvector corresponding to the smaller eigenvalue. (4 Points)

14. **Ex 14: Left eigenvectors and conservation of total stock**

Use again

$$A = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}.$$

(a) Compute A^T. (3 Points)

(b) Find a left eigenvector of A for $\lambda = 1$, equivalently a right eigenvector of A^T for $\lambda = 1$. (5 Points)

(c) Find a left eigenvector of A for $\lambda = 0.5$, equivalently a right eigenvector of A^T for $\lambda = 0.5$. (6 Points)

(d) Show that

$$w = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

describes conservation of total stock, because $w^T x = x_1 + x_2$. (4 Points)

(e) Explain why right and left eigenvectors are generally different when $A \neq A^T$. (2 Points)

15. **Ex 15: Diagonalization and powers of a redistribution matrix**

Use the matrix

$$A = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}.$$

From the previous exercise, one may use the right eigenvectors

$$v_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \qquad v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix},$$

with eigenvalues

$$\lambda_1 = 1, \qquad \lambda_2 = 0.5.$$

(a) Form the matrices

$$P = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix}, \qquad D = \begin{pmatrix} 1 & 0 \\ 0 & 0.5 \end{pmatrix}.$$

Compute P^{-1}. (5 Points)

(b) Verify that

$$A = PDP^{-1}.$$

(5 Points)

(c) Use

$$A^k = PD^kP^{-1}$$

to derive a formula for $A^k x_0$ when

$$x_0 = \begin{pmatrix} 100 \\ 0 \end{pmatrix}.$$

(6 Points)

(d) Compute x_1, x_3, and the limiting stock vector as $k \to \infty$. (4 Points)

16. **Ex 16: Eigenanalysis of a non-symmetric road-relation matrix**

Consider the asymmetric road-relation matrix

$$R = \begin{pmatrix} 0 & 12 \\ 17 & 0 \end{pmatrix}.$$

This is not a stochastic redistribution matrix; it is an algebraic representation of a directed relation.

(a) Show that $R \neq R^T$. (3 Points)

(b) Compute the characteristic equation and determine the eigenvalues. (6 Points)

(c) For each eigenvalue λ, show that

$$v = \begin{pmatrix} 12 \\ \lambda \end{pmatrix}$$

is a right eigenvector. (5 Points)

(d) For the positive eigenvalue, compute one left eigenvector by solving

$$R^T w = \lambda w.$$

Compare it with the right eigenvector. (4 Points)

(e) Explain why asymmetric logistics relations generally have different left and right eigenvectors. (2 Points)

17. **Ex 17: Long-run distribution and decay of imbalance**

A second two-hub redistribution process is given by

$$x_{k+1} = Cx_k, \qquad C = \begin{pmatrix} 0.6 & 0.3 \\ 0.4 & 0.7 \end{pmatrix}.$$

(a) Compute the eigenvalues of C. (5 Points)

(b) Compute right eigenvectors for both eigenvalues. (5 Points)

(c) Decompose

$$x_0 = \begin{pmatrix} 70 \\ 70 \end{pmatrix}$$

into the eigenvector basis of C. (5 Points)

(d) Derive $x_k = C^k x_0$ and compute the limiting vector as $k \to \infty$. (3 Points)

(e) Determine the smallest integer k for which the absolute coefficient of the decaying imbalance mode is below 1. (2 Points)

18. **Ex 18: Modal decomposition of a three-region logistics system**

A logistics update matrix has the following eigenvectors:

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \qquad v_2 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \qquad v_3 = \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix},$$

with corresponding eigenvalues

$$\lambda_1 = 1, \qquad \lambda_2 = 0.6, \qquad \lambda_3 = 0.2.$$

The initial stock vector is given in modal form:

$$x_0 = 100v_1 + 30v_2 - 10v_3.$$

(a) Compute the physical stock vector x_0 in ordinary coordinates. (5 Points)

(b) Write the general expression for

$$x_k = A^k x_0$$

using the modal decomposition. (5 Points)

(c) Compute x_2. (5 Points)

(d) Identify which mode persists, which mode decays faster, and what this means operationally for regional stock balancing. (5 Points)

19. **Ex 19: A matrix that is not diagonalizable**

Consider

$$F = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

(a) Compute the characteristic polynomial of F. (4 Points)

(b) Determine the eigenvalue and the corresponding eigenspace. (5 Points)

(c) Explain why F is not diagonalizable over $\mathbb{R}$. (5 Points)

(d) Show by induction or direct multiplication that

$$F^k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}.$$

(4 Points)

(e) Give a logistics interpretation of a repeated process where a coupling term accumulates linearly even though the only eigenvalue is 1. (2 Points)

20. **Ex 20: Comparing two redistribution designs by eigenvalues**

Two possible redistribution designs for two hubs are

$$A_1 = \begin{pmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 0.6 & 0.3 \\ 0.4 & 0.7 \end{pmatrix}.$$

Both matrices act by

$$x_{k+1} = Ax_k.$$

(a) Verify that both matrices are column-stochastic. What does this imply for total stock? (4 Points)

(b) Compute the eigenvalues of A_1 and A_2. (6 Points)

(c) Compute a right eigenvector for $\lambda = 1$ for each matrix. Interpret the resulting long-run stock ratio. (5 Points)

(d) For the initial vector

$$x_0 = \begin{pmatrix} 100 \\ 0 \end{pmatrix},$$

which design converges faster to its long-run ratio? Justify your answer using the magnitude of the second eigenvalue. (3 Points)

(e) Explain briefly why faster mixing is not always automatically better in logistics. Mention at least one trade-off. (2 Points)